

Practice Problem Set 3

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1. Nonlinear Production and Cost Optimization

A firm produces output according to:

$$Q(L) = 50L^{0.6}e^{-0.02L}$$

where L is labour.

1. Determine the value of L^* that maximizes $Q(L)$.
2. Verify that the second-order condition holds.
3. Interpret economically what happens to output beyond L^* .
4. Compute the elasticity of Q with respect to L at the optimum.

2. Consumer Utility with Exponential Preferences

A consumer's utility function is given by:

$$U(x) = 5x^{0.4}e^{-0.1x}$$

1. Find x^* that maximizes utility.
2. Show that utility first increases and then decreases with x .
3. Compute the maximum utility value $U(x^*)$.
4. Explain intuitively why the exponential term reflects satiation.

3. Logarithmic Profit Maximization

A firm's profit function is:

$$\pi(Q) = 200 \ln(Q + 1) - 30Q$$

1. Derive the first-order condition for maximization.
2. Solve for the optimal output Q^* .
3. Verify that the second-order condition holds.
4. Compute the profit at Q^* .
5. Interpret how logarithmic revenue affects marginal profit.

4. Advertising Response with Diminishing Returns

Sales as a function of advertising A are:

$$S(A) = 400(1 - e^{-0.05A}) - 2A$$

1. Derive the optimal advertising level A^* .
2. Check that the solution satisfies the second-order condition.
3. Compute the corresponding maximum sales $S(A^*)$.
4. Discuss the meaning of $e^{-0.05A}$ in capturing saturation effects.

5. Elasticity via Logarithmic Differentiation

The cost of production is given by:

$$C(Q) = 10Q^{0.7}(2 + \ln Q)$$

1. Use logarithmic differentiation to find the elasticity of cost with respect to output.
2. Evaluate the elasticity when $Q = 10$.
3. Interpret the result: is cost increasing at an increasing or decreasing rate?

6. Maximization with Rational Function

The firm's revenue is modeled by:

$$R(P) = \frac{100P}{4 + P^2}$$

where P is the price per unit.

1. Find the price P^* that maximizes total revenue.
2. Verify that it is indeed a maximum.
3. Determine the corresponding revenue $R(P^*)$.
4. Explain the trade-off between price and quantity implied by this function.

7. Cost Minimization with Logarithmic Form

A firm's cost function is:

$$C(x) = 20x + 50 \ln(x)$$

for $x > 0$.

1. Find the value of x^* that minimizes cost.
2. Confirm that it is a minimum.
3. Interpret the economic meaning of the logarithmic term in cost.
4. Compute the elasticity of cost at x^* .

8. Growth Model with Continuous Time

A population grows according to:

$$P(t) = 800e^{0.03t} - 400e^{-0.02t}$$

1. Find the time t^* at which the population grows fastest.
2. Compute the maximum growth rate.
3. Interpret the combination of exponential growth and decay components.
4. Explain why t^* corresponds to a turning point in $P(t)$.

9. Logarithmic Demand and Revenue

The demand curve is:

$$P(Q) = 60 - 5 \ln(Q + 1)$$

and total revenue is $R(Q) = P(Q)Q$.

1. Derive the first-order condition for maximizing revenue.
2. Solve for the revenue-maximizing quantity Q^* .
3. Verify that $R''(Q^*) < 0$.
4. Discuss the role of logarithmic demand in moderating price sensitivity.

10. Challenge: Nonlinear Utility and Elasticity

A utility function is given by:

$$U(x) = \ln(3x + 2) - 0.5x^2$$

1. Determine the value of x^* that maximizes utility.
2. Check that $U''(x^*) < 0$.
3. Compute the elasticity of utility with respect to x at the optimum.
4. Provide an economic interpretation: why does marginal utility turn negative for large x ?